

1. What is Type Casting?

Type casting means converting a value from **one data type to another data type**.

For example:

```
int x = 10;  
double y = x;
```

Here, **int** is converted to **double**.

```
int → double
```

2. Types of Type Casting

There are two basic types:

1. **Implicit Type Casting**
2. **Explicit Type Casting**

3. Implicit Type Casting

Implicit casting happens automatically.

The compiler performs the conversion without being told.

Example

```
int x = 10;  
double y = x;  
  
cout << y;
```

Output

```
10
```

Here:

```
int → double
```

The compiler automatically converts **10** into **10.0**.

Remember

Implicit = Automatic

4. Explicit Type Casting

Explicit casting is done by the programmer.

The programmer tells C++ which data type to convert the value into.

Example

```
double x = 10.75;  
int y = (int)x;  
  
cout << y;
```

Output

10

Here:

double → int

The decimal part **.75** is removed.

Remember

Explicit = Programmer tells C++

5. Another Example

```
int a = 5;  
int b = 2;  
  
double result = (double)a / b;  
  
cout << result;
```

Output

2.5

The programmer explicitly converts `a`:

`int → double`

So the calculation becomes:

`5.0 / 2 = 2.5`

6. Implicit vs Explicit Casting

Implicit Casting	Explicit Casting
Automatic	Done by programmer
Compiler performs conversion	Programmer specifies conversion
No special casting syntax	Casting syntax is used
<code>double y = x;</code>	<code>int y = (int)x;</code>
Example: <code>int → double</code>	Example: <code>double → int</code>

7. Exmaples

Example 1: Implicit Casting — `int` to `double`

```
#include <iostream>
using namespace std;

int main() {
    int num = 10;
    double result = num;

    cout << result;

    return 0;
}
```

Output:

10

Understand the Code:

- `num` is an `int`.
 - `result` is a `double`.
 - C++ automatically converts `int` to `double`.
 - This is **implicit casting**.
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Example 2: Explicit Casting — `double` to `int`

```
#include <iostream>
using namespace std;

int main() {
    double num = 15.75;
    int result = (int)num;

    cout << result;

    return 0;
}
```

Output:

15

Understand the Code:

- `num` contains `15.75`.
 - `(int)num` converts it to an integer.
 - The decimal part `.75` is removed.
 - This is **explicit casting**.
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Example 3: Explicit Casting in Division

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;
    int b = 2;

    double result = (double)a / b;

    cout << result;
}
```

```
    return 0;  
}
```

Output:

2.5

Understand the Code:

- Normally, $5 / 2$ gives 2 because both values are integers.
- `(double)a` converts 5 into 5.0 .
- Now the calculation becomes $5.0 / 2$.
- The result is 2.5 .

 Remember

Implicit casting: C++ does it automatically.

Explicit casting: The programmer does it using:

```
(type)value
```

Example:

```
(double)a  
(int)num
```